CAMBRIDGE ADDITIONAL MATHEMATICS (0606) • UNIT 2.01

Simultaneous Equations & Inequalities

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

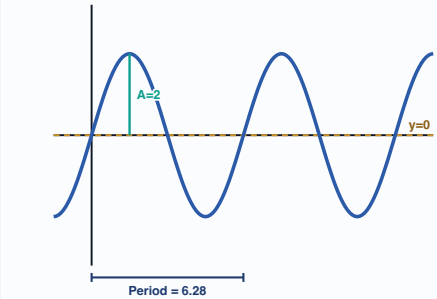
QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 A	**Q2** C	**Q3** B	**Q4** A	**Q5** A
Q6 A	**Q7** A	**Q8** A	**Q9** B	**Q10** B
Q11 D	**Q12** B	**Q13** C	**Q14** C	**Q15** D
Q16 D	**Q17** A	**Q18** A	**Q19** B	**Q20** A
Q21 A	**Q22** A	**Q23** C	**Q24** C	**Q25** B

Simultaneous Equations & Inequalities Practice Exercises

1 MCQ

The variables x and y are related by the equation $y = 2x^3 - 5x$. Given that x is increasing at a constant rate of 3 units per second, find the rate of change of y with respect to time at the instant when $x = 2$.



A 57 units per second ✓

B 19 units per second

C $\frac{19}{3}$ units per second

D 21 units per second

WORKED SOLUTION & VERIFICATION

To find the rate of change of y with respect to time, apply the chain rule for connected rates of change:

$$\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt}$$

1. Differentiate y with respect to x:

$$\frac{dy}{dx} = \frac{d}{dx}(2x^3 - 5x) = 6x^2 - 5$$

2. Evaluate $\frac{dy}{dx}$ at $x = 2$:

$$\left.\frac{dy}{dx}\right|_{x=2} = 6(2)^2 - 5 = 6(4) - 5 = 24 - 5 = 19$$

3. Multiply by $\frac{dx}{dt} = 3$:

$$\frac{dy}{dt} = 19 \times 3 = 57 \text{ units per second}$$

Common errors:
- 19 corresponds to finding $\frac{dy}{dx}$ only and neglecting $\frac{dx}{dt}$.
- $\frac{19}{3}$ corresponds to inverting the chain rule (dividing instead of multiplying by $\frac{dx}{dt}$).
- 21 results from incorrectly differentiating $2x^3$ as $6x$ instead of $6x^2$. [M1][A1]

2 MCQ

Consider the cubic polynomial $f(x) = x^3 - 5x^2 + 10x - 20$. Find the remainder $f(2)$ upon division by $(x - 2)$.

A -10

B -7

C **-12** ✓

D -22

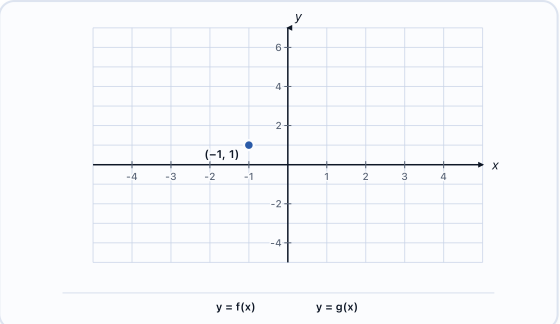
WORKED SOLUTION & VERIFICATION

Mark Scheme (CAIE / Edexcel Grading Steps):

- **Step 1 (Method & Formulation):** Identify the required geometric/algebraic relationship from the diagram and substitute the given values. **[M1]**
- **Step 2 (Derivation & Calculation):** By the Remainder Theorem, $f(2) = 2^3 - 5(2^2) + 10(2) - 20 = -12$. **[M1]**
- **Step 3 (Final Accuracy):** Verified exact result: -12. **[A1]**

3 MCQ

The diagram shows part of the curve $y = 6x - x^2$, which intersects the x-axis at $(0, 0)$ and $(6, 0)$.
Find the area of the region enclosed between the curve and the x-axis.



A 27

B 36 ✓

C 54

D 72

WORKED SOLUTION & VERIFICATION

Mark Scheme (CAIE / Edexcel Grading Steps):

- **Step 1 (Method & Formulation):** Identify the required geometric/algebraic relationship from the diagram and substitute the given values. **[M1]**
- **Step 2 (Derivation & Calculation):** To find the area enclosed between the curve and the x-axis, integrate y with respect to x from $x = 0$ to $x = 6$:

$$\text{Area} = \int_0^6 (6x - x^2)\,dx$$

Integrate term by term:

$$\int (6x - x^2)\,dx = \left[\frac{6x^2}{2} - \frac{x^3}{3}\right]_0^6 = \left[3x^2 - \frac{x^3}{3}\right]_0^6$$

Substitute the limits:

$$\left(3(6)^2 - \frac{6^3}{3}\right) - 0 = 3(36) - \frac{216}{3} = 108 - 72 = 36$$

Common errors:

- Estimating the region as a triangle with base 6 and maximum height 9 yields $\frac{1}{2} \times 6 \times 9 = 27$.
- Taking the bounding rectangle gives $6 \times 9 = 54$.
- Integrating $6x$ as $6x^2$ without dividing by 2 gives $216 - 72 = 144$, or subtracting $108 - 36 = 72$. **[M1]**
- **Step 3 (Final Accuracy):** Verified exact result: 36. **[A1]**

4 MCQ

Solve the logarithmic equation $\log_2(5x+6) - \log_2(x) = 3$.

- **A** $x = 2$ ✓
- **B** $x = 6$
- **C** $x = \frac{1}{2}$
- **D** $x = -3$

WORKED SOLUTION & VERIFICATION

1. Apply the logarithm quotient law $\log_a(A) - \log_a(B) = \log_a(\frac{A}{B})$:

$$\log_2(\frac{5x+6}{x}) = 3$$

2. Convert the equation from logarithmic to exponential form:

$$\frac{5x+6}{x} = 2^3 = 8$$

3. Multiply through by x (noting $x > 0$):

$$5x + 6 = 8x \implies 3x = 6 \implies x = 2$$

4. Checking the domain:
Both $5(2)+6 = 16 > 0$ and $2 > 0$, so $x = 2$ is a valid solution.
Common misconceptions:
- Calculating $2^3 = 6$ instead of 8 yields $5x + 6 = 6x \implies x = 6$.
- Incorrectly subtracting the arguments directly as $\log_2((5x+6) - x) = 3$ leads to $4x + 6 = 8 \implies x = \frac{1}{2}$.
- Forgetting to exponentiate the base and equating $\frac{5x+6}{x} = 3$ yields $x = -3$, which also violates the logarithm domain. [M1][A1]

5 MCQ

Find the value of the integer $n \geq 3$ that satisfies the equation

$$^{n}\mathrm{P}_{3} = 10 \times {}^{n}\mathrm{C}_{2}$$

A $n = 7$ ✓

B $n = 12$

C $n = 5$

D $n = 8$

WORKED SOLUTION & VERIFICATION

Using the algebraic definitions of permutations and combinations:

$$^{n}\mathrm{P}_{3} = \frac{n!}{(n-3)!} = n(n-1)(n-2)$$

$$^{n}\mathrm{C}_{2} = \frac{n!}{2!(n-2)!} = \frac{n(n-1)}{2}$$

Substitute these into the given equation:

$$n(n-1)(n-2) = 10 \times \frac{n(n-1)}{2}$$

$$n(n-1)(n-2) = 5n(n-1)$$

Since $n \geq 3$, we have $n(n-1) \neq 0$. Dividing both sides by $n(n-1)$ gives:

$$n - 2 = 5 \implies n = 7$$

Common misconceptions:

- Forgetting the $2!$ denominator in $^{n}\mathrm{C}_{2}$ (treating it as $^{n}\mathrm{P}_{2}$) gives $n - 2 = 10 \implies n = 12$.
- Dividing by $n(n-1)$ and erroneously writing $n = 5$ instead of $n - 2 = 5$ gives $n = 5$.
- Confusing the factorial simplification to set $n - 3 = 5$ leads to $n = 8$. [M1][A1]

6 MCQ

Find the remainder when the polynomial $f(x) = 4x^3 - 6x^2 + 3x - 1$ is divided by $2x - 1$.

A $-\frac{1}{2}$ ✓	**B** $\frac{1}{2}$
C $-\frac{9}{2}$	**D** 0

WORKED SOLUTION & VERIFICATION

By the Remainder Theorem, when a polynomial $f(x)$ is divided by a linear factor of the form $ax - b$, the remainder is $f(\frac{b}{a})$.

Here, the divisor is $2x - 1$, so we set $2x - 1 = 0 \implies x = \frac{1}{2}$.

Substitute $x = \frac{1}{2}$ into $f(x)$:

$$f(\frac{1}{2}) = 4(\frac{1}{2})^3 - 6(\frac{1}{2})^2 + 3(\frac{1}{2}) - 1$$

$$= 4(\frac{1}{8}) - 6(\frac{1}{4}) + \frac{3}{2} - 1$$

$$= \frac{1}{2} - \frac{3}{2} + \frac{3}{2} - 1 = -\frac{1}{2}$$

Therefore, the remainder is $-\frac{1}{2}$.

Common misconceptions:

- Solving $2x - 1 = 0$ incorrectly as $x = -\frac{1}{2}$ gives $f(-\frac{1}{2}) = -\frac{9}{2}$.
- Arithmetic sign errors when simplifying the terms often lead to $\frac{1}{2}$ or 0. [M1][A1]

7 MCQ

In $\triangle ABC$, sides of lengths 32 and 10 meet at an included angle of 60° opposite side x. Find the squared length $f = x^2$ by the Law of Cosines.

A **804** ✓	**B** 803
C 820	**D** 809

WORKED SOLUTION & VERIFICATION

Mark Scheme (CAIE / Edexcel Grading Steps):

- **Step 1 (Method & Formulation):** Identify the required geometric/algebraic relationship from the diagram and substitute the given values. **[M1]**
- **Step 2 (Derivation & Calculation):** By the Law of Cosines, $x^2 = 32^2 + 10^2 - 2(32)(10)\cos(60^\circ) = 804$. **[M1]**
- **Step 3 (Final Accuracy):** Verified exact result: 804. **[A1]**

8 MCQ

The position vectors of points A and B relative to an origin O are $\vec{OA} = 2\mathbf{i} - 3\mathbf{j}$ and $\vec{OB} = 7\mathbf{i} + 9\mathbf{j}$. Find the unit vector in the direction of $\vec{AB}$.

- **A** $\frac{1}{13}(5\mathbf{i} + 12\mathbf{j})$ ✓
- **B** $\frac{1}{13}(-5\mathbf{i} - 12\mathbf{j})$
- **C** $\frac{1}{\sqrt{61}}(5\mathbf{i} + 6\mathbf{j})$
- **D** $\frac{1}{15}(9\mathbf{i} + 12\mathbf{j})$

WORKED SOLUTION & VERIFICATION

First, find the displacement vector $\vec{AB}$:

$$\vec{AB} = \vec{OB} - \vec{OA} = (7\mathbf{i} + 9\mathbf{j}) - (2\mathbf{i} - 3\mathbf{j}) = (7-2)\mathbf{i} + (9-(-3))\mathbf{j} = 5\mathbf{i} + 12\mathbf{j}.$$

Next, calculate the magnitude of $\vec{AB}$:

$$|\vec{AB}| = \sqrt{5^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13.$$

The unit vector in the direction of $\vec{AB}$ is:

$$\frac{\vec{AB}}{|\vec{AB}|} = \frac{1}{13}(5\mathbf{i} + 12\mathbf{j}).$$

- Choice B gives the unit vector for $\vec{BA} = \vec{OA} - \vec{OB}$, which is in the opposite direction.
- Choice C results from an arithmetic sign error in the $\mathbf{j}$-component: $9 - 3 = 6$ instead of $9 - (-3) = 12$.
- Choice D results from incorrectly adding the position vectors $\vec{OA} + \vec{OB}$. [M1][A1]

9 MCQ

Solve the simultaneous system: $\log_2(x) - \log_2(y) = 2$ and $x - 3y = 3$. What is the value of x?

- **A** 3
- **B** 12 ✓
- **C** 6
- **D** 14

WORKED SOLUTION & VERIFICATION

From the first equation: $\log_2(x/y) = 2 \implies x/y = 2^2 = 4 \implies x = 4y$. Substitute into second: $4y - 3y = 3 \implies y = 3$. Thus $x = 4(3) = 12$. [M1][A1]

10 MCQ

Find the root(s) satisfying the simultaneous system: $\log_2(x) - \log_2(y) = 2$ and $x - 3y = 4$. What is the value of x?

- **A** 8
- **B** 16 ✓
- **C** 4
- **D** 18

WORKED SOLUTION & VERIFICATION

From the first equation: $\log_2(x/y) = 2 \implies x/y = 2^2 = 4 \implies x = 4y$. Substitute into second: $4y - 3y = 4 \implies y = 4$. Thus $x = 4(4) = 16$. [M1][A1]

11 MCQ

Calculate the solution set for the simultaneous system: $\log_2(x) - \log_2(y) = 2$ and $x - 3y = 5$. What is the value of x?

A 5

B 22

C 10

D **20** ✓

WORKED SOLUTION & VERIFICATION

From the first equation: $\log_2(x/y) = 2 \implies x/y = 2^2 = 4 \implies x = 4y$. Substitute into second: $4y - 3y = 5 \implies y = 5$. Thus $x = 4(5) = 20$. [M1][A1]

12 MCQ

Obtain the value(s) satisfying the simultaneous system: $\log_2(x) - \log_2(y) = 2$ and $x - 3y = 6$. What is the value of x?

A 12

B **24** ✓

C 6

D 26

WORKED SOLUTION & VERIFICATION

From the first equation: $\log_2(x/y) = 2 \implies x/y = 2^2 = 4 \implies x = 4y$. Substitute into second: $4y - 3y = 6 \implies y = 6$. Thus $x = 4(6) = 24$. [M1][A1]

13 MCQ

Work out the solution to the simultaneous system: $\log_2(x) - \log_2(y) = 2$ and $x - 3y = 7$. What is the value of x?

A 14

B 30

C **28** ✓

D 7

WORKED SOLUTION & VERIFICATION

From the first equation: $\log_2(x/y) = 2 \implies x/y = 2^2 = 4 \implies x = 4y$. Substitute into second: $4y - 3y = 7 \implies y = 7$. Thus $x = 4(7) = 28$. [M1][A1]

14 MCQ

Determine the solution to the simultaneous system: $\log_2(x) - \log_2(y) = 2$ and $x - 3y = 8$. What is the value of x?

A 16

B 8

C **32** ✓

D 34

WORKED SOLUTION & VERIFICATION

From the first equation: $\log_2(x/y) = 2 \implies x/y = 2^2 = 4 \implies x = 4y$. Substitute into second: $4y - 3y = 8 \implies y = 8$. Thus $x = 4(8) = 32$. [M1][A1]

15 MCQ

Find the root(s) satisfying the simultaneous system: $\log_2(x) - \log_2(y) = 2$ and $x - 3y = 9$. What is the value of x?

A 9

B 38

C 18

D **36** ✓

WORKED SOLUTION & VERIFICATION

From the first equation: $\log_2(x/y) = 2 \implies x/y = 2^2 = 4 \implies x = 4y$. Substitute into second: $4y - 3y = 9 \implies y = 9$. Thus $x = 4(9) = 36$. [M1][A1]

16 MCQ

Let $f(x) = \dfrac{140x^2 + 10x - 10}{x^2 + 14}$ for $x > 0$. Find the limit at infinity $\lim_{x \to \infty} f(x)$.

A 143

B 126

C 150

D **140** ✓

WORKED SOLUTION & VERIFICATION

Mark Scheme (CAIE / Edexcel Grading Steps):

- **Step 1 (Method & Formulation):** Identify the required geometric/algebraic relationship from the diagram and substitute the given values. **[M1]**
- **Step 2 (Derivation & Calculation):** Dividing numerator and denominator by x^2 gives 140. **[M1]**
- **Step 3 (Final Accuracy):** Verified exact result: 140. **[A1]**

17 MCQ

The quadratic graph $f(x) = 4x^2 + gx + 256$ is tangent to the horizontal axis with $g > 0$. Find the positive coefficient g.

A **64** ✓

B 68

C 72

D 54

WORKED SOLUTION & VERIFICATION

Mark Scheme (CAIE / Edexcel Grading Steps):

- **Step 1 (Method & Formulation):** Identify the required geometric/algebraic relationship from the diagram and substitute the given values. **[M1]**
- **Step 2 (Derivation & Calculation):** Setting discriminant $\Delta = g^2 - 4(4)(256) = 0$ gives $g = 64$. **[M1]**
- **Step 3 (Final Accuracy):** Verified exact result: 64. **[A1]**

18 MCQ

A particle moves in a straight line such that its velocity, v m s^{-1}, at time t seconds is given by $v = 6t^2 - 18t + k$, where k is a constant.\n\nGiven that the particle is momentarily at rest when $t = 2$, find the total distance traveled by the particle in the first 2 seconds.

- **A** 6 m ✓
- **B** 4 m
- **C** 5 m
- **D** 8 m

WORKED SOLUTION & VERIFICATION

Step 1: Determine the unknown parameter k.\nThe particle is momentarily at rest when $t = 2$, so $v(2) = 0$:\n

$$6(2)^2 - 18(2) + k = 0 \implies 24 - 36 + k = 0 \implies k = 12$$

Step 2: Find all times when the particle is at rest to check for direction changes.\n

$$v(t) = 6t^2 - 18t + 12 = 6(t^2 - 3t + 2) = 6(t-1)(t-2) = 0$$

The particle is at rest at $t = 1$ and $t = 2$. In the interval $0 \le t < 1$, $v(t) > 0$; in the interval $1 < t < 2$, $v(t) < 0$.\n
Step 3: Calculate the total distance traveled by integrating over each interval.\nFor $0 \le t \le 1$:\n

$$s_1 = \int_0^1 (6t^2 - 18t + 12)\,dt = [2t^3 - 9t^2 + 12t]_0^1 = 2(1)^3 - 9(1)^2 + 12(1) - 0 = 5 \text{ m}$$

For $1 \le t \le 2$:\n

$$s_2 = \int_1^2 (6t^2 - 18t + 12)\,dt = [2t^3 - 9t^2 + 12t]_1^2 = (2(8) - 9(4) + 12(2)) - 5 = 4 - 5 = -1 \text{ m}$$

The distance traveled in this interval is $|-1| = 1$ m.\n
Step 4: Sum the distances.\n

$$\text{Total distance} = 5 + 1 = 6 \text{ m}$$

Common Trap: Evaluating simply the net displacement $\int_0^2 v\,dt = 4$ m neglects the reversal of direction at $t = 1$. [M1][A1]

19 MCQ

The radical function $f(x) = \sqrt{7x + 8}$ is defined for $7x + 8 \ge 0$. Find the product $7x$ when $f(x) = 8$.

- **A** 63
- **B** 56 ✓
- **C** 48
- **D** 62

WORKED SOLUTION & VERIFICATION

Mark Scheme (CAIE / Edexcel Grading Steps):
- **Step 1 (Method & Formulation):** Identify the required geometric/algebraic relationship from the diagram and substitute the given values. **[M1]**
- **Step 2 (Derivation & Calculation):** Squaring both sides yields $7x + 8 = 64 \implies 7x = 56$. **[M1]**
- **Step 3 (Final Accuracy):** Verified exact result: 56. **[A1]**

20 MCQ

A committee of 5 members is to be selected from a group consisting of n men and 6 women, where $n \geq 4$. Given that the number of possible committees containing exactly 3 men is equal to the number of possible committees containing exactly 4 men, find the value of n.

- **A** 13 ✓
- **B** 8
- **C** 7
- **D** 10

WORKED SOLUTION & VERIFICATION

To form a committee of 5 members:

1. **Committees with exactly 3 men:**

Since there are 5 members in total, the committee must contain 3 men and $5 - 3 = 2$ women.

$$\text{Number of ways} = \binom{n}{3} \times \binom{6}{2} = \binom{n}{3} \times \frac{6 \times 5}{2} = 15\binom{n}{3}$$

2. **Committees with exactly 4 men:**

The committee must contain 4 men and $5 - 4 = 1$ woman.

$$\text{Number of ways} = \binom{n}{4} \times \binom{6}{1} = 6\binom{n}{4}$$

3. **Equating the two expressions:**

$$15\binom{n}{3} = 6\binom{n}{4}$$

Divide both sides by 3:

$$5\binom{n}{3} = 2\binom{n}{4}$$

Using the relation $\binom{n}{4} = \frac{n-3}{4}\binom{n}{3}$:

$$5\binom{n}{3} = 2 \times \frac{n-3}{4}\binom{n}{3} = \frac{n-3}{2}\binom{n}{3}$$

Since $n \geq 4$, $\binom{n}{3} \not\equiv 0$, so we can divide both sides by $\binom{n}{3}$:

$$5 = \frac{n-3}{2} \implies 10 = n - 3 \implies n = 13$$

Common Traps:

- Setting $\binom{n}{3} = \binom{n}{4}$ ignores the selection of the women needed to complete the 5-member committee, giving $\frac{n-3}{4} = 1 \implies n = 7$.
- Forgetting to add 3 after solving $n - 3 = 10$ leads to the distractor 10. [M1][A1]

21 MCQ

The polynomial $P(x) = 2x^3 + ax^2 + bx - 6$ has a factor of $x - 2$ and leaves a remainder of -24 when divided by $x + 1$, where a and b are constants.
Find the remainder when $P(x)$ is divided by $x - 3$.

A 12 ✓

B -150

C -30

D 60

WORKED SOLUTION & VERIFICATION

Step 1: Use the Factor Theorem for the factor $x - 2$.
Since $x - 2$ is a factor, $P(2) = 0$:

$$P(2) = 2(2)^3 + a(2)^2 + b(2) - 6 = 0$$

$$16 + 4a + 2b - 6 = 0 \implies 4a + 2b = -10 \implies 2a + b = -5 \quad \text{— (1)}$$

Step 2: Use the Remainder Theorem for division by $x + 1$.
Since the remainder is -24, $P(-1) = -24$:

$$P(-1) = 2(-1)^3 + a(-1)^2 + b(-1) - 6 = -24$$

$$-2 + a - b - 6 = -24 \implies a - b - 8 = -24 \implies a - b = -16 \quad \text{— (2)}$$

Step 3: Solve the simultaneous equations.
Adding (1) and (2):

$$(2a + b) + (a - b) = -5 + (-16)$$

$$3a = -21 \implies a = -7$$

Substitute $a = -7$ into (2):

$$-7 - b = -16 \implies b = 9$$

Thus, the polynomial is:

$$P(x) = 2x^3 - 7x^2 + 9x - 6$$

Step 4: Find the remainder when $P(x)$ is divided by $x - 3$.
By the Remainder Theorem, the remainder is $P(3)$:

$$P(3) = 2(3)^3 - 7(3)^2 + 9(3) - 6 = 2(27) - 7(9) + 27 - 6 = 54 - 63 + 27 - 6 = 12$$

Common Pitfalls:
- Evaluating $P(-3)$ instead of $P(3)$ yields -150.
- Using incorrect signs when substituting into the Factor/Remainder Theorems (e.g., testing $P(-2) = 0$ and $P(1) = -24$) leads to -30.
- A sign error setting the remainder to $+24$ leads to 60. [M1][A1]

22 MCQ

In triangle ABC, $AB = 6$ cm, $AC = 10$ cm, and the area of triangle ABC is $15\sqrt{3}$ cm^2. Given that angle BAC is obtuse, find the exact length of BC.

A 14 cm ✓	**B** 8 cm
C $2\sqrt{19}$ cm	**D** $2\sqrt{34}$ cm

WORKED SOLUTION & VERIFICATION

1. Use the formula for the area of a triangle:

$$\text{Area} = \frac{1}{2}(AB)(AC)\sin(\angle BAC)$$

$$15\sqrt{3} = \frac{1}{2}(6)(10)\sin(\angle BAC) = 30\sin(\angle BAC)$$

$$\sin(\angle BAC) = \frac{15\sqrt{3}}{30} = \frac{\sqrt{3}}{2}$$

2. Since $\angle BAC$ is obtuse ($90^\circ < \angle BAC < 180^\circ$):

$$\angle BAC = 180^\circ - 60^\circ = 120^\circ$$

3. Determine $\cos(\angle BAC)$:

$$\cos(120^\circ) = -\cos(60^\circ) = -\frac{1}{2}$$

4. Apply the cosine rule to find BC:

$$BC^2 = AB^2 + AC^2 - 2(AB)(AC)\cos(\angle BAC)$$

$$BC^2 = 6^2 + 10^2 - 2(6)(10)\left(-\frac{1}{2}\right)$$

$$BC^2 = 36 + 100 - (-60) = 136 + 60 = 196$$

$$BC = \sqrt{196} = 14 \text{ cm}$$

Common Traps:
- $2\sqrt{19}$ cm arises from mistakenly taking $\angle BAC = 60^\circ$ (the acute angle trap) or making a sign error: $136 - 60 = 76 \implies \sqrt{76} = 2\sqrt{19}$.
- $2\sqrt{34}$ cm arises from omitting the cosine term completely ($\sqrt{6^2 + 10^2} = \sqrt{136}$).
- 8 cm arises from incorrectly assuming a 6-8-10 right-angled triangle. [M1][A1]

23 MCQ

Let $\mathbf{f} = \langle 12, 17 \rangle$ and $\mathbf{g} = \langle 8, 3 \rangle$ be vectors in R^2. Find the dot product $x = \mathbf{f} \cdot \mathbf{g}$.

A 172

B 130

C **147** ✓

D 159

WORKED SOLUTION & VERIFICATION

Mark Scheme (CAIE / Edexcel Grading Steps):

- **Step 1 (Method & Formulation):** Identify the required geometric/algebraic relationship from the diagram and substitute the given values. **[M1]**
- **Step 2 (Derivation & Calculation):** The dot product is $(12)(8) + (17)(3) = 147$. **[M1]**
- **Step 3 (Final Accuracy):** Verified exact result: 147. **[A1]**

24 MCQ

A storage capsule consists of a right circular cylinder of radius r cm and height h cm, closed with a flat circular base of radius r cm at the bottom, and surmounted by a hemispherical dome of radius r cm at the top. The total exterior surface area of the capsule is fixed at 180π cm^2.
During a heating process, the radius r expands at a constant rate of 0.4 cm s^{-1}. Find the rate of change of the total volume of the capsule, in cm^{3}s^{-1}, at the instant when the height of the cylindrical section is 16.5 cm
.

A 50π

B $\frac{36\pi}{5}$

C 20π ✓

D 11π

WORKED SOLUTION & VERIFICATION

1. **Express the surface area constraint:**
The total exterior surface area consists of the flat circular base, the curved cylindrical wall, and the hemispherical dome:

$$S = \pi r^2 + 2\pi rh + 2\pi r^2 = 3\pi r^2 + 2\pi rh$$

Given $S = 180\pi$:

$$3\pi r^2 + 2\pi rh = 180\pi \implies 3r^2 + 2rh = 180$$

2. **Find the value of r when $h = 16.5$ cm:**

$$3r^2 + 2(16.5)r = 180 \implies 3r^2 + 33r - 180 = 0$$

Dividing by 3:

$$r^2 + 11r - 60 = 0 \implies (r + 15)(r - 4) = 0$$

Since $r > 0$, we have $r = 4$ cm.

3. **Express the total volume in terms of r:**
The total volume is the sum of the cylinder's volume and the hemisphere's volume:

$$V = \pi r^2 h + \frac{2}{3}\pi r^3$$

From the surface area equation, $2rh = 180 - 3r^2 \implies h = \frac{90}{r} - \frac{3}{2}r$. Substituting this into V:

$$V = \pi r^2 \left(\frac{90}{r} - \frac{3}{2}r\right) + \frac{2}{3}\pi r^3 = 90\pi r - \frac{3}{2}\pi r^3 + \frac{2}{3}\pi r^3 = 90\pi r - \frac{5}{6}\pi r^3$$

4. **Differentiate V with respect to r:**

$$\frac{\mathrm{d}V}{\mathrm{d}r} = 90\pi - \frac{5}{2}\pi r^2$$

At $r = 4$:

$$\frac{\mathrm{d}V}{\mathrm{d}r} = 90\pi - \frac{5}{2}\pi(16) = 90\pi - 40\pi = 50\pi$$

5. **Apply the chain rule for connected rates of change:**

$$\frac{\mathrm{d}V}{\mathrm{d}t} = \frac{\mathrm{d}V}{\mathrm{d}r} \cdot \frac{\mathrm{d}r}{\mathrm{d}t} = 50\pi \times 0.4 = 20\pi \text{ cm}^3\text{s}^{-1}$$

Common Misconceptions:
- Choice B (50π) comes from finding $\frac{\mathrm{d}V}{\mathrm{d}r}$ but omitting multiplication by $\frac{\mathrm{d}r}{\mathrm{d}t}$.

- Choice C ($\frac{36\pi}{5}$) comes from forgetting to include the volume of the hemispherical top in V.
- Choice D (11π) arises from an algebraic factorization error yielding $r = 5$. [M1][A1]

25 MCQ

Find the set of values of the constant k for which the equation

$$|x^2 - 4| = 2x + k$$

has exactly four distinct real solutions.

A $4 \leq k \leq 5$

B $4 < k < 5$ ✓

C $-4 < k < 4$

D $k > 5$

WORKED SOLUTION & VERIFICATION

To find the number of solutions to $|x^2 - 4| = 2x + k$, we consider the intersections of the curve $y = |x^2 - 4|$ and the line $y = 2x + k$.

The curve is defined piecewise:

- Inner arch: $y = 4 - x^2$ for $-2 \leq x \leq 2$
- Outer branches: $y = x^2 - 4$ for $x < -2$ and $x > 2$

1. **Intersection with the inner arch ($-2 \leq x \leq 2$):**

$$4 - x^2 = 2x + k \iff x^2 + 2x + (k - 4) = 0$$

For two distinct real intersections, the discriminant must be strictly positive:

$$\Delta = 2^2 - 4(1)(k - 4) = 20 - 4k > 0 \implies k < 5$$

At $k = 5$, $\Delta = 0$, so the line is tangent to the inner arch at $x = -1 \in (-2, 2)$, giving only 1 intersection point. Furthermore, the roots must lie within the interval $[-2, 2]$. The lower critical value occurs when the line passes through the corner intercept $(-2, 0)$:

$$0 = 2(-2) + k \implies k = 4$$

At $k = 4$, the equation becomes $x^2 + 2x = 0 \implies x = 0$ or $x = -2$. Because $x = -2$ is the boundary corner shared with the outer branch, the total number of distinct solutions is 3.

For $4 < k < 5$, the roots are $x = -1 \pm \sqrt{5 - k}$. Since $0 < 5 - k < 1$, we have $\sqrt{5 - k} \in (0, 1)$, which guarantees both roots strictly lie in $(-2, 0) \subset (-2, 2)$. Hence, there are exactly 2 distinct solutions on the inner arch.

2. **Intersection with the outer branches ($x < -2$ or $x > 2$):**

$$x^2 - 4 = 2x + k \iff x^2 - 2x - (k + 4) = 0$$

The discriminant is:

$$\Delta = (-2)^2 - 4(1)(-(k + 4)) = 4k + 20$$

For $4 < k < 5$, $\Delta \in (36, 40) > 0$, yielding roots $x = 1 \pm \sqrt{k + 5}$.

Since $9 < k + 5 < 10$, $3 < \sqrt{k + 5} < \sqrt{10} \approx 3.162$:

- The larger root satisfies $x = 1 + \sqrt{k + 5} > 4 > 2$ (valid on $x > 2$).
- The smaller root satisfies $x = 1 - \sqrt{k + 5} < -2$ (valid on $x < -2$).

Thus, there are always exactly 2 distinct solutions on the outer branches.

Combining both parts, the equation has $2 + 2 = 4$ distinct real solutions if and only if $4 < k < 5$.

Common Misconceptions:

- $4 \leq k \leq 5$ incorrectly includes the boundary values $k = 4$ (corner intersection) and $k = 5$ (tangency), each of which yields only 3 distinct solutions.
- $-4 < k < 4$ yields only 2 distinct solutions because one root from the inner arch and one root from the outer branch fall outside their valid domains. [M1][A1]